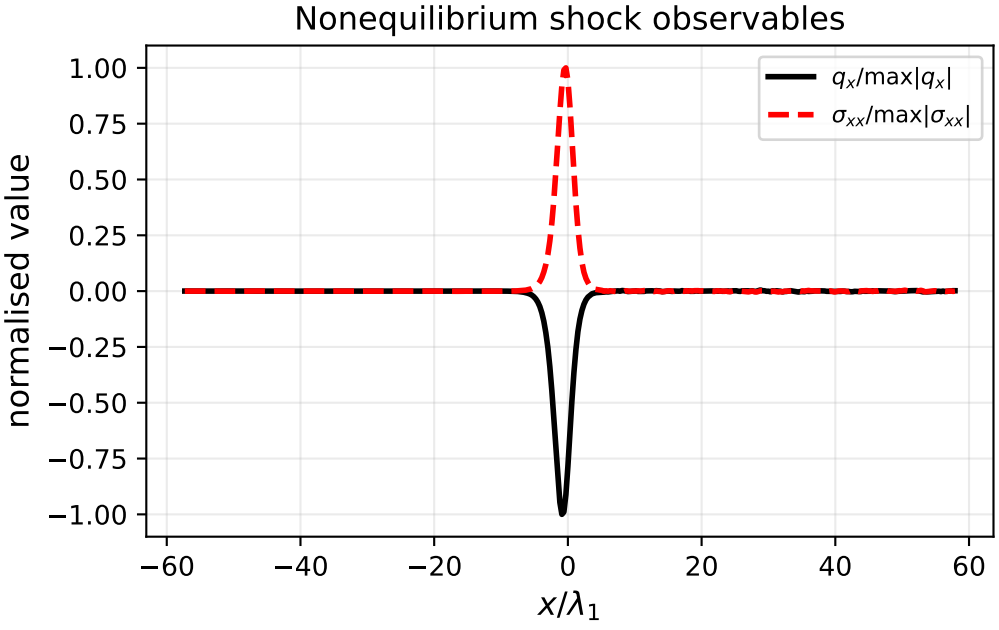
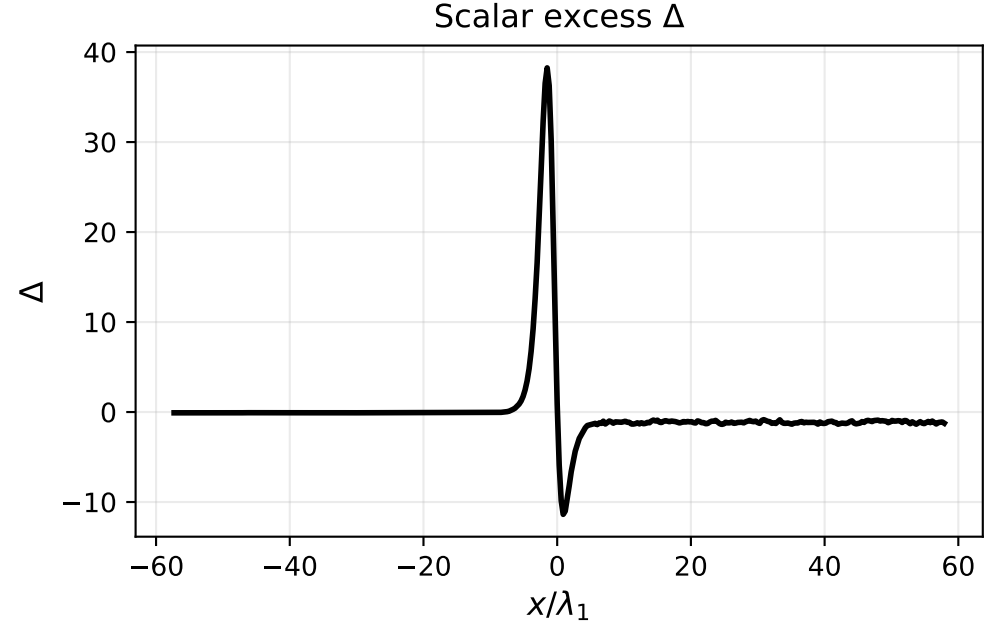
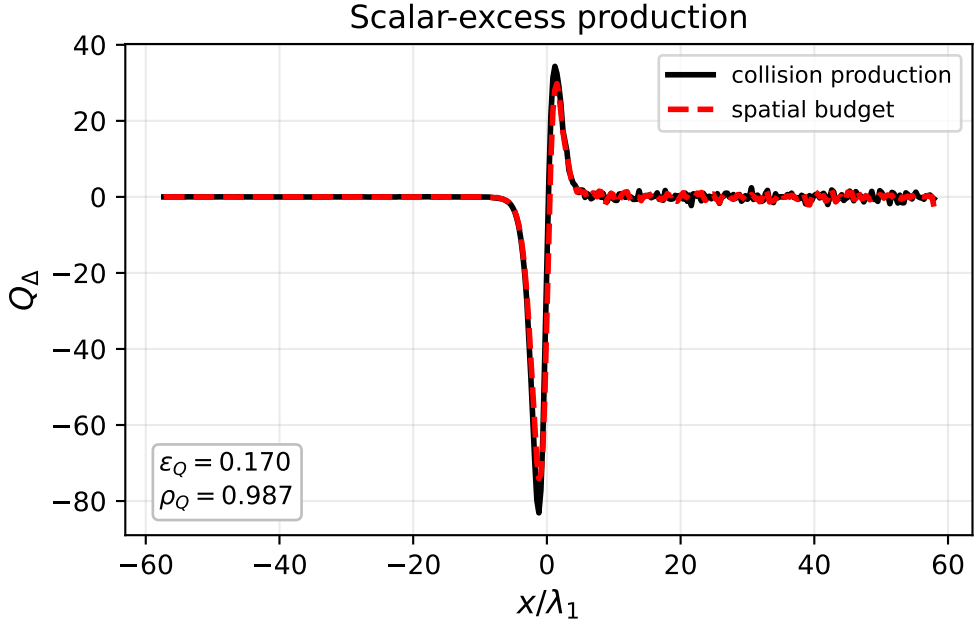

Nonequilibrium shock observables
normalised value
x/λ_1
$q_x/\max|q_x|$
$\sigma_{xx}/\max|\sigma_{xx}|$

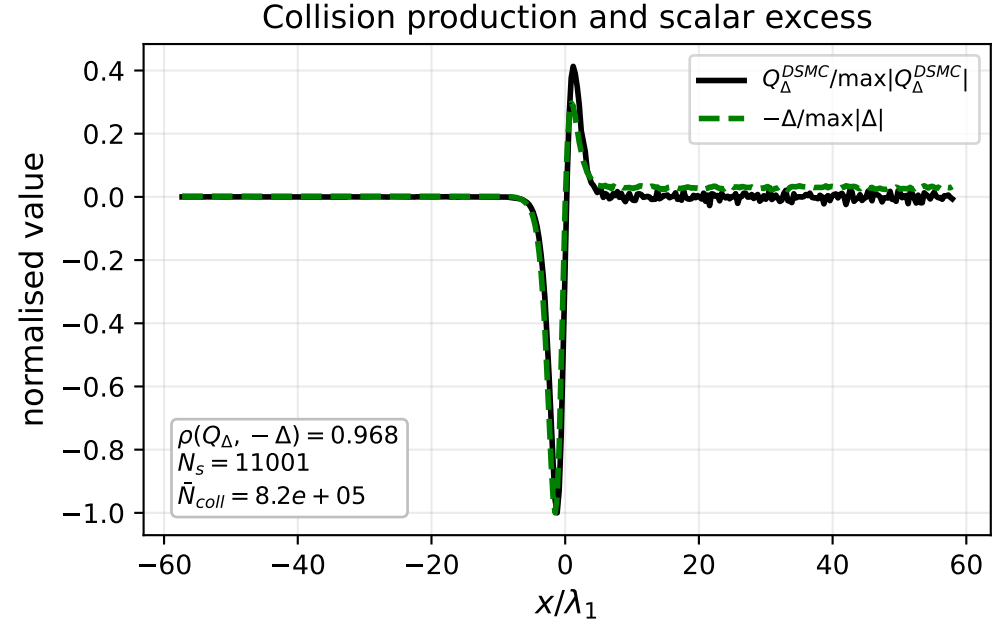
(a)


Scalar excess Δ
Δ
x/λ_1

(b)


Scalar-excess production
Q_Δ
x/λ_1
collision production
spatial budget
$\varepsilon_Q = 0.170$
$\rho_Q = 0.987$

(c)


Collision production and scalar excess
normalised value
x/λ_1
$Q_\Delta^{DSMC}/\max|Q_\Delta^{DSMC}|$
$-\Delta/\max|\Delta|$
$\rho(Q_\Delta, -\Delta) = 0.968$
$N_s = 11001$
$\bar{N}_{coll} = 8.2e+05$

(d)